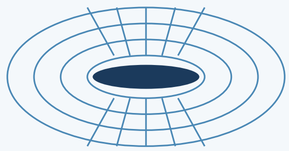
Why HEC-RAS Has No Physics-Validated Meshing

— and why that gap is the opportunity

HEC-RAS Mesh AI · research one-pager · Aug 2026

CFD has refined meshes against physics for decades. Every riverine flood package still meshes by hand. Five reasons — none of them blocks this tool.

CFD — WHERE ADAPTATION PAYS

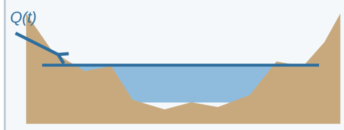


mesh refines toward the geometry & the error

- Geometry fixed & exact (CAD)
- Forcing known — often steady
- One scalar objective (lift, drag)
- Discretization error dominates

ADJOINT ADAPTATION: refine exactly where error moves the answer

RIVERINE FLOOD — WHERE IT NEVER DID

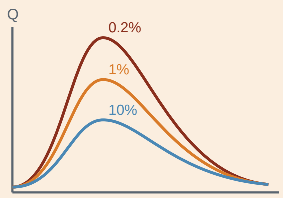


wet/dry edge moves with the event

- Terrain & roughness uncertain
- Forcing = runoff hydrograph
- Output = inundation map × ensemble
- One fixed auditable mesh (FEMA)

NO SINGLE FLOW STATE EXISTS TO ADAPT A MESH TO

THE PRACTITIONER'S POINT (HEC-RAS developer, pers. comm.)



hydrologic variability: the forcing is a statistical construct, not a measurement

"In CFD the input flows are easily modeled. Rainfall-runoff response hydrographs carry hydrologic variability and occur transiently over a time scale."

Variability half: the inflow inherits storm-pattern, moisture and loss-rate uncertainty — refining the mesh below that error floor buys nothing.

Transient half: the flood sweeps through days of states — the “optimal” adapted mesh differs by event and by hour. The wet domain is the answer.

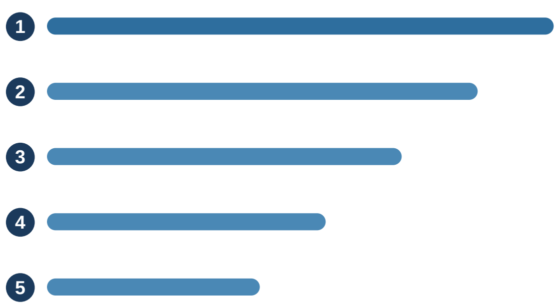
SUB-GRID BATHYMETRY CHANGED WHAT A MESH IS FOR (v5.0, 2015)

ONE coarse cell — fine terrain inside

property tables

Since 2015 each cell carries the full-resolution terrain as precomputed elevation-volume and face-conveyance curves (Casulli 2009). The mesh no longer resolves terrain — only flow gradients. Coarse static cells stay accurate, the convergence curve flattens (TUFLOW SGS: ±10% at 20 m vs a 1 m baseline), and the classic CFD reason to refine disappears.

FIVE REASONS, RANKED BY WEIGHT



Error budget inverted — and calibration launders mesh error
terrain, n and inflow uncertainty dwarf discretization error; calibrated n silently absorbs mesh error (Horritt & Bates 2001; Savage 2016; Bates 2022)

Sub-grid bathymetry removed most of the need
terrain fidelity moved into property tables; static coarse cells stay accurate (Casulli 2009; HEC docs; TUFLOW SGS)

Wrong objective structure — the hydrograph reason
map output × design-event ensemble on one fixed auditable mesh; moving wet/dry domain; no scalar functional for an adjoint (FEMA no-rise: exact reproduction to 0.01 ft)

The solver architecture resists in-run adaptation
near-orthogonal Voronoi cells, no hanging nodes, global implicit solve, property tables invalidated by any mesh change; only TIME is adaptive

Theory gaps + economics + youth
wet/dry fronts break error estimators; ~25-person free gov code vs aerospace-funded meshing vendors; RAS 2D meshing is only ~10 years old

0 of 17

practice-grade riverine 2D flood packages ship solution-adaptive meshing (survey, 2026)

Tsunami ≠ rivers

AMR reached practice only with huge scale separation: GeoClaw, NTHMP-approved 2012. Riverine floods are slow & broadly wet at peak

Literature gap

no published grid-convergence method exists for sub-grid solvers — Stage 6 of this project builds one

THE OPENING THE FIVE REASONS LEAVE — exactly this tool

OFFLINE

Mesh design happens between runs via the automation harness — no in-solver surgery, no property-table regeneration mid-run.

LEARNED

Trained on expert breakline practice + terrain; answers the one question sub-grid leaves open: where flow gradients need resolution.

PHYSICS-SCORED

Every mesh ships with a quantitative error-vector certificate vs a refined reference — output stays one static, regulator-compatible mesh.

Sources: HEC-RAS 2D User's & Technical Reference Manuals · Bates, Annu. Rev. Fluid Mech. 2022 · Horritt & Bates 2001 · Savage et al. 2016 · Casulli 2009 · EA SC120002 · FEMA 2D Recommended Practices 2021 · GeoClaw/NTHMP · Kesserwani & Sharifian 2023 · full citations: docs/research/why-no-physics-validated-meshing.md